

Representation-Centric Taxonomy

29 works • 9 top-level regimes (R0–R8)
R4a/R4b are experimental subregimes of one top-level R4

REGIME	SHORT NAME	REPRESENTATION FORM	FIRST VISIBILITY	CONSUMER / DECISION REACH	EMPIRICAL STATUS
R0	Opaque callback	Call site; internals hidden	Run time	Runtime / execution only	Strong empirical
R1	Scalar metadata	Cost / selectivity scalar	Costing	Cost model / cost-only	Strong empirical
R2	Analytical / learned summary	Input-conditioned behavior	Analytical	Cost estimator / work estimate	Partial empirical
R3	Adaptive / profiled summary	Profiles and feedback	Post-execution	Estimator / reoptimizer	Artifact-blocked
R4a	Full/native relational exposure	Relational tree Froid / PostgreSQL	Pre-plan rewrite	Rewriter + enumerator path / join / pushdown	Strong empirical
R4b	Selective relational exposure	Selective/residual PRISM	Pre-plan rewrite	Rewriter + enumerator selective plan reach	Partial empirical
R5	Static semantics / properties	Dependencies and effects	Pre-plan analysis	Rewriter + enumerator safety / placement	Literature-only
R6	Execution / dataflow IR	Compiler + plan IR QFusor	Post-plan	Execution optimizer fusion / vectorization	Literature-only
R7	Source-aware learned structure	Source CFG + SQL plan GRACEFUL	Costing	Learned estimator cost / placement model	Partial empirical
R8	Cross-layer synthesis	$S(U \mid Q, p)$ hybrid evidence	Proposed	Multiple consumers interface requirements	Synthesis + components

STATUS KEY



Strong empirical



Partial empirical



Literature-only



Artifact-blocked



Synthesis + components

UDFBench is a cross-regime benchmark anchor, not a representation regime.

Status describes this project's evidence—not field-wide maturity.